Summation and Singularities under Polylogarithmic Reversion

Convergent angular reconstruction, sharp Lambert late terms, and removable inverse endpoints

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Abstract

We investigate three ways in which inversion changes a polylogarithmic expansion. First, the harmonic depth-two angular-zero transseries is known to diverge even after equal exponential bases are collected. We construct a specified summation of its original multiset coefficient family: sum each nonlinear degree at subunit radius, take its radial Abel limit, and then sum the degree blocks. For every real $a \geq 6$, $b > 0$, and $0 < \rho \leq 1$, this last series converges absolutely to the actual angular displacement, with a uniform explicit geometric error bound. At integer orders all its coefficients belong to one fixed finitely generated algebra of polylogarithm values. A finite frequency-and-degree truncation attains $\exp(-\lambda a^2 + O_\lambda(a))$ error for every $0 < \lambda < \log(3/2)$.

Second, we prove the precise even/odd late-coefficient conjecture in the new resonant Lambert inverse report. Uniform factorial convolution estimates control the previously unjustified summation over the number of factors. We obtain a first correction, a universal leading-amplitude law for analytic inverse cores, and an exact initial-disk Borel decomposition into two logarithms and two dilogarithms with a continuously differentiable remainder. Both nearest Borel endpoints are singular for every nonzero complex coupling.

Third, we classify the minimal endpoint ramification of the direct inverse of Li_s for every real $s > 1$. Its endpoint is holomorphic exactly at $s = 1 + 1/q$, $q \geq 2$ integral. We give its convergent Taylor charts and disprove a proposed all-real-order extension of an integer-order cut-modulus inequality. Classical ingredients and previous repository results are attributed; proofs, finite exact checks, and ordinary numerical diagnostics retain distinct scopes.

Keywords. Polylogarithm; harmonic sums; transseries; Lagrange inversion; Abel summation; factorial asymptotics; Borel transform; ramification; Lambert series.

Contents

1 Questions, conclusions, and provenance **3**
1.1 The three research targets . . . 3
1.2 Relation to established theory . . . 3
1.3 Conventions and evidence . . . 4

2 A complex neighborhood for the harmonic angular equation **4**

3 A convergent summation of the angular multiset transseries **6**
3.1 The exact summation order . . . 7
3.2 Why divergence persists in the original ordering . . . 8

4 Finite polylogarithm coordinates for every degree **8**

5 A moving finite truncation with Gaussian error **9**
5.1 An exact complex frequency remainder . . . 10

6 Sharp late coefficients of a resonant Lambert inverse **11**
6.1 The source equation and its normalization . . . 11
6.2 Analytic-core universality . . . 12
6.3 The first correction and the analytic data it detects . . . 14
6.4 Exact logarithm–dilogarithm decomposition in the Borel disk . . . 15

7 Exact ramification of the inverse at the polylogarithm endpoint **16**
7.1 Taylor charts at the removable endpoints . . . 18
7.2 A correction to a proposed endpoint-circle extension . . . 20

8 Verification, interpretation, and integration **20**
8.1 Exact finite checks . . . 21
8.2 Angular numerical diagnostics . . . 21
8.3 Lambert and endpoint diagnostics . . . 22
8.4 Recommended changes to the existing material . . . 23

9 Further research questions **23**
9.1 Angular summation and finite approximation . . . 23
9.2 Factorial reversion and Borel geometry . . . 24
9.3 Global inverse geometry . . . 24

1 Questions, conclusions, and provenance

An inverse expansion has several distinct mathematical meanings. A formal coefficient formula can be valid without its sum converging. A convergent analytic representation can require a different grouping from the original asymptotic scale. The nearest singularities of a Borel transform depend on exact late coefficients, while a branch point of the forward function can disappear from its inverse. This article addresses these distinctions in concrete problems drawn from the intersection of ProveIt's Polylogarithms and Transseries projects.

The source baseline is the commit displayed on the title page, inspected on 12 October 2026. In particular, the three then-current incoming archives were read before selecting the research targets. The article does not treat earlier unanswered questions as unanswered without checking those arrivals. The accompanying provenance file records exact repository paths and archive members, so that the contribution remains reviewable after the reports are moved into their canonical locations.

1.1 The three research targets

The angular-zero report [5] supplies an explicit multiset-indexed expansion with an error after every fixed finite exponential cutoff. Its continuation [6] proves that the complete expansion diverges at every fixed integer outer order, even after equal rational bases are combined. Research question 14 of that continuation asks for a justified summation method. We answer it for all real $a \geq 6$ by an Abel limit within each fixed nonlinear degree, followed by an ordinary convergent sum over degrees (Theorem 3.1). The estimate comes from an actual holomorphic norm on a fixed complex disk, not from differentiation of a boundary Fourier series beyond its convergence range.

Question 13 of the same continuation asks for optimal moving truncation. We give a rigorously controlled finite approximation with an error that is Gaussian in the large outer order (Theorem 5.2). This is a constructive upper bound for a specified double cutoff. We do not identify the sharp optimum for truncation in the original exponential-base ordering.

The incoming resonant Lambert inverse report [8] proves the exact inverse coefficient formula and its sharp Borel radius for positive real coupling. Equations `eq:conj-even` and `eq:conj-odd` there explicitly conjecture the leading hyperbolic amplitudes. Section 6 proves them with a locally uniform complex-parameter remainder and one further term. The estimates are uniform in the number of Lagrange factors, closing the exact gap identified by that report.

Finally, the convergent noninteger charts in [9, `prop:noninteger`] permit a sharper local question: what is the *smallest* branched cover required by the inverse at $w = \zeta(s)$? Theorem 7.1 answers it for every real $s > 1$. The exceptional holomorphic family also clarifies why the real-order endpoint geometry cannot be inferred from the integer-order theorem.

1.2 Relation to established theory

Lagrange inversion, including the auxiliary-parameter version used here, is classical; Gessel's survey [1] gives its formal versions and proofs. Our use of Rouché's theorem supplies the additional analytic domain and quantitative bounds needed for the angular summation. The positive depth-two kernel and global angular-zero uniqueness are already repository results [7]; we rederive the kernel moments because their uniform estimates are central to the new argument.

The general theory of factorially divergent series is also established. Borinsky [2], building on earlier work including Bender, develops product, composition, and inverse rules for such series. Our elementary convolution proof is consistent with that framework. Its role is to prove the specific two-action Lambert conjecture, retain uniformity in the complex coupling, and make

the first correction and initial-disk Borel regularity explicit. Neither factorial reversion itself nor the general existence of resurgent Lambert expansions is claimed as new. For the latter, see Dorigoni–Kleinschmidt [3].

The direct inverse classification uses the classical convergent Jonquière expansion [4]. It sharpens the repository's existing two-variable charts; the general mechanism of Puiseux uniformization is classical. The newest quadratic-feedback arrival [10] contains related minimal-ramification calculations for a different implicit function. We credit this nearby precedent and distinguish the direct inverse studied here.

Accordingly, the novelty claim is specific: these are proved answers and refinements relative to the inspected corpus, supported by a targeted primary-source search. No exhaustive worldwide priority claim is made for every corollary or special-case identity.

1.3 Conventions and evidence

All logarithms are continued from their explicitly specified real branches. The shifted Borel transform is

$$\mathcal{B}\left(\sum_{n\geq 1} c_n q^n\right)(\xi) = \sum_{n\geq 1} \frac{c_n}{\Gamma(n)} \xi^{n-1}.$$

An estimate locally uniform in u means uniform on every compact subset of the stated parameter domain. Statements about a holomorphic function inside a disk and continuous boundary values do not imply analytic continuation across the boundary.

Every theorem below has a conventional proof. The supplied programs check finite algebraic identities and evaluate independent numerical diagnostics. The decimals are not interval enclosures, and the programs are not proof-assistant formalizations. This manuscript was prepared with AI assistance; it is intended for mathematical review before incorporation into a collective manuscript.

2 A complex neighborhood for the harmonic angular equation

For real $a, b > 0$ define, initially in the unit disk,

$$F_{a,b}(z) = \sum_{n\geq 2} \frac{H_{n-1}^{(b)}}{n^a} z^n, \qquad H_m^{(b)} = \sum_{j=1}^{m} j^{-b}. \tag{2.1}$$

This is the depth-two polylogarithm $\mathrm{Li}_{a,b}(z,1)$ when the multiple-polylogarithm convention uses $n > m \geq 1$ and the factor z^n. The elementary inequality $1 \leq H_m^{(b)} \leq m$ will be used uniformly over all $b > 0$.

Lemma 2.1 (Positive kernel and exact moments)**.** *On* $\mathbb{C} \setminus [1, \infty)$*,*

$$F_{a,b}(z) = z^2 \int_{0<y<x<1} \frac{W_{a,b}(x,y)}{(1-xz)(1-yz)}\, dy\, dx, \quad W_{a,b}(x,y) = \frac{(-\log x)^{a-1}(\log(x/y))^{b-1}}{\Gamma(a)\Gamma(b)}. \tag{2.2}$$

For nonnegative integers j, k *its moments are*

$$\int_{0<y<x<1} W_{a,b}(x,y) x^j y^k\, dy\, dx = \frac{1}{(j+k+2)^a (k+1)^b}. \tag{2.3}$$

Proof. Set $x = e^{-t}$ and $y = e^{-t-v}$. The Jacobian is xy, and the integral in (2.3) separates into Gamma integrals with decay rates $j+k+2$ and $k+1$. Expanding both geometric denominators

in (2.2), the coefficient of z^n is

$$\sum_{k=0}^{n-2} \frac{1}{n^a(k+1)^b} = \frac{H_{n-1}^{(b)}}{n^a}.$$

Absolute convergence justifies this computation for $|z| < 1$. On every compact subset of the slit plane the denominators are bounded away from zero, and W has finite mass 2^{-a}. Dominated convergence and differentiation on compact sets give the holomorphic continuation. This is the kernel of [7], with its moments made explicit for later estimates. □

For $a \geq 6$, $b > 0$, and $0 < \rho \leq 1$, let $\theta_{a,b}(\rho)$ be the unique zero in $(0, \pi)$ of $\operatorname{Im} F_{a,b}(\rho e^{i\theta})$. In the range needed here its existence and simplicity follow from the global angular theorem of [7]; alternatively, the construction below independently gives the root near $\pi/2$. Write

$$\delta_{a,b}(\rho) = \frac{\pi}{2} - \theta_{a,b}(\rho).$$

The Gaussian sign in the same source puts this displacement in $(0, \pi/2)$. We shall also see its local positivity directly.

Set

$$\begin{aligned} R_{a,b}(z) &= F_{a,b}(z) - 2^{-a} z^2, \\ z_+(v) &= i\rho e^{-iv}, \qquad z_-(v) = -i\rho e^{iv}, \\ S(v) &= \frac{2^a}{2i\rho^2}\big(R_{a,b}(z_+(v)) - R_{a,b}(z_-(v))\big), \qquad (2.4) \\ B(v) &= \frac{v}{\sin 2v}, \qquad B(0) = \frac{1}{2}, \qquad f(v) = -B(v)S(v). \qquad (2.5) \end{aligned}$$

These definitions are holomorphic; the difference in (2.4) is used instead of taking an imaginary part at complex v. For real v it equals the normalized imaginary tail. The angular equation is exactly

$$\sin 2v + S(v) = 0. \qquad (2.6)$$

Lemma 2.2 (Uniform analytic tail bound)**.** *Let $0 < r < \pi/2$ and $c = \cos r$. On the disk $|v| \leq r$,*

$$|S(v)| \leq \rho e^{3r}\big(c^{-2} + 2^{-b}c^{-1}\big)(2/3)^a. \qquad (2.7)$$

At $r = 1/4$ this implies

$$\sup_{|v| \leq 1/4} |f(v)| < \frac{1}{4}\eta_{a,\rho}, \qquad \eta_{a,\rho} = 10\rho(2/3)^a. \qquad (2.8)$$

In particular $a \geq 6$ gives $\eta_{a,\rho} \leq 640/729 < 1$.

Proof. If z lies in either sector of half-width r about i or $-i$, then $|1 - tz| \geq \cos r = c$ for every $t \geq 0$: this is the distance from 1 to a ray whose angle is at least $\pi/2 - r$ in absolute value. The elementary identity

$$\frac{1}{(1-A)(1-C)} - 1 = \frac{A}{(1-A)(1-C)} + \frac{C}{1-C}$$

and the moments of Wx and Wy give

$$|R_{a,b}(z)| \leq |z|^3 3^{-a}\big(c^{-2} + 2^{-b}c^{-1}\big).$$

For $|v| \leq r$, both $z_\pm(v)$ lie in these sectors and have modulus at most ρe^r. Substitution proves (2.7).

For completeness the constants can be bounded rationally:

$$\cos(1/4) > 31/32, \qquad e^{3/4} < 17/8, \qquad \sup_{|v|\le 1/4} |B(v)| < 8/15.$$

The last inequality follows from $|\sin(2v)/(2v) - 1| \le \sinh(1/2)/(1/2) - 1 < 1/16$. For that strict bound the positive tail is at most $(1/24)/(1 - 1/80) = 10/237 < 1/16$. Taylor's formula and a geometric bound on its positive tail give $e^{3/4} < 18427/8704 < 17/8$; the cosine bound follows from $\cos x > 1 - x^2/2$. Consequently

$$4 \sup |B| e^{3/4}(c^{-2} + c^{-1}) < \frac{45696}{4805} < 10.$$

This proves (2.8), including its strictness. □

3 A convergent summation of the angular multiset transseries

The scalar parameter used below scales the complete perturbation before taking boundary values. It is not an independent variation of infinitely many Fourier coefficients. Define the coupled equation

$$\delta(\tau) = \tau f(\delta(\tau)), \qquad \delta(0) = 0. \tag{3.1}$$

Theorem 3.1 (Convergent angular reconstruction)**.** *For real $a \ge 6$, $b > 0$, and $0 < \rho \le 1$, put*

$$D_k(a, b; \rho) = \frac{1}{k}[v^{k-1}] f(v)^k = \frac{(-1)^k}{k}[v^{k-1}] B(v)^k S(v)^k. \tag{3.2}$$

Then

$$\delta_{a,b}(\rho) = \sum_{k \ge 1} D_k(a, b; \rho) \tag{3.3}$$

converges absolutely. More precisely,

$$|D_k(a, b; \rho)| \le \frac{\eta_{a,\rho}^k}{4k}, \tag{3.4}$$

$$\left| \delta_{a,b}(\rho) - \sum_{k=1}^{K} D_k(a, b; \rho) \right| \le \frac{\eta_{a,\rho}^{K+1}}{4(K+1)(1 - \eta_{a,\rho})}, \qquad K \ge 0. \tag{3.5}$$

The solution of (3.1) is holomorphic on $|\tau| < \eta_{a,\rho}^{-1}$, which contains $\tau = 1$.

Proof. On $|v| = 1/4$, Lemma 2.2 gives $|\tau f(v)| < |v|$ when $|\tau| \eta_{a,\rho} < 1$. Rouché's theorem shows that $v - \tau f(v)$ has exactly one zero, counted with multiplicity, in the disk. It is therefore simple. The implicit function theorem and uniqueness patch its local branches into one holomorphic function of τ on the indicated disk. Lagrange inversion gives (3.2). Applying Cauchy's coefficient estimate to $f(v)^k$ on the fixed circle gives

$$|D_k| \le \frac{1}{k}(1/4)^{1-k}(\eta_{a,\rho}/4)^k = \frac{\eta_{a,\rho}^k}{4k}.$$

At real τ the zero is real by conjugation symmetry. Moreover, $S(0) < 0$. Indeed the imaginary part of the positive kernel at $z = i\rho$ is strictly negative: $z^2 = -\rho^2$, while the imaginary part of $((1 - i\rho x)(1 - i\rho y))^{-1}$ is strictly positive for $0 < y < x < 1$. The removed quadratic term is real. Thus $f(0) > 0$, the zero starts positive for small positive τ, and it cannot cross zero for any $\tau > 0$. At $\tau = 1$ it is in $(0, 1/4)$ and solves the actual angular equation. Global angular uniqueness identifies it with the stated $\delta_{a,b}(\rho)$. Finally summing $\sum_{k>K} \eta^k/k \le \eta^{K+1}/((K+1)(1-\eta))$ proves the error bound. □

3.1 The exact summation order

A multiset of Fourier modes is a finitely supported sequence $\boldsymbol{m} = (m_3, m_4, \ldots)$ of nonnegative integers. Set

$$k = |\boldsymbol{m}| = \sum m_n, \quad \Lambda(\boldsymbol{m}) = \prod_{n\geq 3} (n/2)^{m_n}, \quad R(\boldsymbol{m}) = \sum_{n\geq 3} (n-2)m_n, \quad h_n = H_{n-1}^{(b)}.$$

The source coefficient is

$$C(\boldsymbol{m}) = \frac{(-1)^k (k-1)!}{\prod m_n!} [v^{k-1}] B(v)^k \prod_{n\geq 3} \sin(n\pi/2 - nv)^{m_n}. \tag{3.6}$$

The source proves that the sum with $\Lambda < B_0$, for fixed $B_0 > 1$, approximates the displacement with error $O_{B_0}(\rho B_0^{-a})$ as $a \to \infty$ [5]. The support is locally finite in Λ: every participating n is less than $2B_0$ and $k < \log B_0 / \log(3/2)$.

Theorem 3.2 (Degree blocks and their Abel boundary values)**.** *For $0 < \rho < 1$ and fixed k, the following inner sum converges absolutely:*

$$D_k(a, b; \rho) = \sum_{|\boldsymbol{m}|=k} C(\boldsymbol{m}) \rho^{R(\boldsymbol{m})} \prod h_n^{m_n} \Lambda(\boldsymbol{m})^{-a}. \tag{3.7}$$

For $a \geq 6$, $b > 0$, its radial boundary values satisfy

$$\begin{aligned} D_k(a, b; 1) &= \lim_{\rho \uparrow 1} \sum_{|\boldsymbol{m}|=k} C(\boldsymbol{m}) \rho^{R(\boldsymbol{m})} \prod h_n^{m_n} \Lambda(\boldsymbol{m})^{-a}, \\ \delta_{a,b}(1) &= \sum_{k\geq 1} D_k(a, b; 1). \end{aligned} \tag{3.8}$$

The final sum is ordinary and absolutely convergent, with (3.5) at $\rho = 1$.

Proof. For subunit ρ, on a sufficiently small complex disk $|v| < \epsilon < -\log \rho$, the Fourier expansion

$$S(v) = \sum_{n\geq 3} h_n \rho^{n-2} (2/n)^a \sin(n\pi/2 - nv)$$

and every fixed number of its derivatives converge absolutely. For fixed k one may expand its kth power and extract the coefficient of degree $k-1$ absolutely. The multinomial coefficient and the factor $1/k$ give exactly (3.6).

For $\rho \uparrow 1$, the kernel formula gives uniform convergence of S on $|v| \leq 1/4$: the denominators have a common nonzero lower bound and the numerator is dominated by a constant times the integrable weight. Each fixed Taylor coefficient in (3.2) therefore converges. The summable majorant $\eta_{a,1}^k/(4k)$ is independent of ρ, so dominated convergence also interchanges the radial limit with the outer degree sum. This proves all the assertions with the displayed order of operations. □

Remark 3.3 (Which data the summation uses)*.* This is a summation of the canonical multiset family, retaining both $|\boldsymbol{m}|$ and $R(\boldsymbol{m})$. Equal rational Λ can arise with different degrees and different radial weights. The prescription is therefore not asserted to be a summation functional depending only on the already grouped scalar coefficients $\widehat{C}(\Lambda; b)$. All arithmetic collisions are retained in (3.7); they are not discarded.

3.2 Why divergence persists in the original ordering

For clarity, the divergence proof in [6] is recalled in a short form. It also extends verbatim from integer a to every fixed real $a > 0$. Choose an integer s with $2s > a$. The multiset with $2s$ copies of mode 3 and one copy of an odd prime $P \neq 3$ has exponential base $q_P = 2^{2s+1}/(3^{2s}P)$. This base has a unique multiset representation. If another representation has k factors, then

$$\prod_{j=1}^{k} n_j = 2^{k-(2s+1)} 3^{2s} P.$$

Integrality first forces $k \geq 2s+1$. Counting prime factors then forces every n_j to be prime. Since $n_j \geq 3$, no factor two is possible; hence $k = 2s+1$ and the representation is unique.

The coefficient extraction contains

$$[v^{2s}] B(v)^{2s+1} \cos^{2s}(3v) \cos(Pv) = \frac{(-1)^s P^{2s}}{2^{2s+1}(2s)!} + O_s(P^{2s-2}).$$

Its term after multiplication by q_P^a has magnitude a positive constant times $H_{P-1}^{(b)} P^{2s-a}(1 + O(P^{-2}))$, which tends to infinity. Thus even the terms grouped by equal bases fail to tend to zero. In (3.8), by contrast, the inner boundary blocks are Abel limits, not ordinary sums. Absolute convergence of the outer degree series does not assert absolute convergence of the flattened multiset family, even for every subunit ρ.

4 Finite polylogarithm coordinates for every degree

Let $S_j = S^{(j)}(0)$. Differentiating the holomorphic definition (2.4), rather than a boundary Fourier series, yields

$$S_j = \frac{2^a}{2i\rho^2} \Big((-i)^j F_{a-j,b}(i\rho) - i^j F_{a-j,b}(-i\rho) \Big) - 2^j \sin(j\pi/2). \tag{4.1}$$

Lower outer orders here are defined by the analytic identity $z\partial_z F_{s,b} = F_{s-1,b}$. For any finite number of derivatives one can start with a positive outer order and differentiate its slit-plane continuation, so this notation is unambiguous.

The first four degree blocks are

$$\begin{aligned} D_1 &= -\frac{S_0}{2}, \qquad D_2 = \frac{S_0 S_1}{4}, \\ D_3 &= -\frac{S_0^3}{12} - \frac{S_0^2 S_2}{16} - \frac{S_0 S_1^2}{8}, \\ D_4 &= \frac{S_0^3 S_1}{6} + \frac{S_0^3 S_3}{96} + \frac{3 S_0^2 S_1 S_2}{32} + \frac{S_0 S_1^3}{16}. \end{aligned} \tag{4.2}$$

More generally, if $B(v) = \sum \beta_j v^j$, $S(v) = \sum s_j v^j$, and $f_j = -\sum_{\ell=0}^{j} \beta_\ell s_{j-\ell}$, then

$$D_k = \frac{1}{k} \sum_{j_1 + \cdots + j_k = k-1} f_{j_1} \cdots f_{j_k}. \tag{4.3}$$

This is a finite polynomial formula, evaluated by truncated polynomial multiplication in the accompanying program.

Proposition 4.1 (Reduction at nonpositive outer order)**.** *For every integer $m \geq 0$ and real $b > 0$,*

$$F_{-m,b}(z) = \sum_{j=0}^{m} \binom{m}{j} \operatorname{Li}_{j-m}(z) \operatorname{Li}_{b-j}(z) \tag{4.4}$$

on the common slit-plane continuation.

Proof. Summing first over $n > j$ in the unit disk gives $F_{0,b}(z) = \mathrm{Li}_0(z)\,\mathrm{Li}_b(z)$, where $\mathrm{Li}_0(z) = z/(1-z)$. Apply $(z\partial_z)^m$, use Leibniz' rule and $z\partial_z \mathrm{Li}_s = \mathrm{Li}_{s-1}$, and then analytically continue. Each factor with nonpositive integral order is rational. The identity itself is elementary; its role is to close the coordinate algebra of the convergent reconstruction. □

Corollary 4.2 (A fixed period algebra)**.** *For integers $a \geq 6$, $b \geq 1$, every $D_k(a,b;1)$ belongs to the same finitely generated real algebra*

$$\mathbb{Q}\left[\begin{array}{ll} \mathrm{Re}\, F_{r,b}(i), \mathrm{Im}\, F_{r,b}(i), & 1 \leq r \leq a, \\ \mathrm{Re}\,\mathrm{Li}_j(i), \mathrm{Im}\,\mathrm{Li}_j(i), & 1 \leq j \leq b \end{array}\right]. \tag{4.5}$$

Increasing the nonlinear degree introduces no additional depth-two generators or classical-polylogarithm generators.

Proof. Use (4.1), (4.3), and (4.4). All nonpositive integral-order polylogarithms at i belong to $\mathbb{Q}(i)$, and values at $-i$ are complex conjugates. The operations required are finite sums and products with rational coefficients. This proves membership, not minimality or algebraic independence of the generating set. □

5 A moving finite truncation with Gaussian error

The convergent degree reconstruction still uses analytically evaluated polylogarithms in each block. It is useful to have a completely finite alternative involving only a finite number of harmonic numbers and trigonometric Taylor coefficients. Let $M \geq 3$ be an integer and put

$$\begin{aligned} S_M(v) &= \sum_{n=3}^{M} H_{n-1}^{(b)} \rho^{n-2} (2/n)^a \sin(n\pi/2 - nv), \\ f_M(v) &= -B(v) S_M(v), \qquad D_{k,M} = \frac{1}{k}[v^{k-1}] f_M(v)^k, \\ P_{M,K}(a,b;\rho) &= \sum_{k=1}^{K} D_{k,M}. \end{aligned} \tag{5.1}$$

This keeps modes at most M and total degree at most K. It is a finite subfamily of the original multiset formula, with every collision inside that subfamily combined automatically.

Theorem 5.1 (Explicit finite approximation bound)**.** *Let $a \geq 8$, $b > 0$, $0 < \rho \leq 1$, $M \geq 3$, and $K \geq 0$. If*

$$\kappa = 14 M \rho (2/3)^a \leq \frac{1}{2}, \tag{5.2}$$

then

$$|\delta_{a,b}(\rho) - P_{M,K}(a,b;\rho)| \leq \frac{\rho 2^a M^{2-a}}{a-2} + \frac{\kappa^{K+1}}{M(K+1)(1-\kappa)}. \tag{5.3}$$

The two displayed terms control frequency removal and degree removal, respectively.

Proof. Use the disk $|v| \leq 1/M$. Here $|B(v)| \leq 1$ and $|\sin(n\pi/2 - nv)| \leq e^{n/M} \leq e$ for $n \leq M$. For $a \geq 4$,

$$\sum_{n \geq 3} (n-1)(2/n)^a \leq (2/3)^a \, 81 \left(\zeta(3) - \zeta(4) - \frac{1}{16}\right) < 5(2/3)^a.$$

Since $5e < 14$, it follows that $\sup |f_M| < 14\rho(2/3)^a = \kappa/M$. The same Rouché and Lagrange argument on radius $1/M$ gives a unique real small root δ_M and

$$|D_{k,M}| \leq \frac{\kappa^k}{Mk}, \qquad \left|\delta_M - \sum_{k=1}^{K} D_{k,M}\right| \leq \frac{\kappa^{K+1}}{M(K+1)(1-\kappa)}.$$

The actual root is also in this real interval. At its fixed point, Lemma 2.2 gives $|\delta_{a,b}| \le (10/4)\rho(2/3)^a < 1/M$ under (5.2). On $|v| \le 1/M$ the real derivative of the finite equation $E_M(v) = \sin 2v + S_M(v)$ satisfies

$$\begin{aligned} E'_M(v) &\ge 2\cos(2/M) - \sum_{n\ge 3} n(n-1)(2/n)^a \\ &\ge 2\cos(2/3) - 14(2/3)^a > 1. \end{aligned}$$

For the second line, comparison at exponent five uses

$$\sum_{n\ge 3} n(n-1)(3/n)^5 = 243\left(\zeta(3) - \zeta(4) - \frac{1}{16}\right) < 14.$$

This inequality has a finite rational certificate: sum $(n-1)/n^4$ from 3 through 100 and bound the decreasing tail by $1/(2\cdot 100^2) - 1/(3\cdot 100^3)$. Multiplying by 243 gives a rational number less than 14. Finally, $2\cos(2/3) \ge 14/9$ and $14/9 - 14(2/3)^8 = 6622/6561 > 1$.

At the actual root the residual of the finite equation is its removed Fourier tail, whose magnitude is at most

$$\rho 2^a \sum_{n>M} n^{1-a} \le \frac{\rho 2^a M^{2-a}}{a-2}.$$

The derivative lower bound and the mean value theorem control $|\delta_{a,b} - \delta_M|$ by this residual. Adding the degree error proves (5.3). □

Theorem 5.2 (An attainable Gaussian error scale)**.** *Put $\ell = \log(3/2)$. For every fixed $0 < \lambda < \ell$ choose*

$$M(a) = \lfloor e^{\lambda a} \rfloor, \qquad K(a) = \left\lceil \frac{\lambda a}{\ell - \lambda} \right\rceil. \tag{5.4}$$

There are constants $a_\lambda, C_\lambda > 0$, independent of $b > 0$ and $0 < \rho \le 1$, such that for $a \ge a_\lambda$,

$$|\delta_{a,b}(\rho) - P_{M(a),K(a)}(a, b; \rho)| \le \exp(-\lambda a^2 + C_\lambda a). \tag{5.5}$$

Proof. We have $\log M = \lambda a + O(e^{-\lambda a})$ and $\kappa \le 14e^{-(\ell-\lambda)a}$, so the hypotheses of Theorem 5.1 hold eventually, uniformly in b, ρ. The logarithm of its frequency bound is at most $-\lambda a^2 + (2\lambda + \log 2)a + O(\log a)$. For the degree bound use $K + 1 \ge \lambda a/(\ell - \lambda)$ and $\log \kappa \le \log 14 - (\ell - \lambda)a$. Its logarithm is likewise at most $-\lambda a^2 + O_\lambda(a)$. Increasing C_λ absorbs their sum and all fixed thresholds. □

A simple choice is $\lambda = \ell/2$, giving $M = \lfloor (3/2)^{a/2} \rfloor$ and $K = \lceil a \rceil$. The theorem measures error, not computational optimality: M grows exponentially with a. It is a proved partial answer to the moving truncation question, while the sharp error of a single exponential-base cutoff remains open.

5.1 An exact complex frequency remainder

The positive kernel gives another useful finite identity. Put $h_d(x, y) = \sum_{j=0}^{d} x^{d-j} y^j$. Direct multiplication gives

$$\frac{1}{(1-xz)(1-yz)} - \sum_{d=0}^{M-3} h_d(x, y) z^d = \frac{z^{M-2}(h_{M-2}(x, y) - zxyh_{M-3}(x, y))}{(1-xz)(1-yz)}. \tag{5.6}$$

In the sectors of Lemma 2.2, integration and the exact moments therefore imply

$$\left| F_{a,b}(z) - \sum_{n=2}^{M-1} \frac{H_{n-1}^{(b)}}{n^a} z^n \right| \le \frac{|z|^M}{c^2} \left(\frac{H_{M-1}^{(b)}}{M^a} + |z| \frac{H_{M-1}^{(b)} - 1}{(M+1)^a} \right). \tag{5.7}$$

Unlike an estimate obtained by summing the defining Fourier tail, this remains valid for $|z| > 1$ in those sectors. At fixed complex radius r, its normalized angular version is $O_r(\rho^{M-2}Me^{Mr}(2/M)^a)$. Taking M proportional to a/r yields $\exp(-a\log a + O_r(a))$. The stronger Gaussian construction above instead shrinks the analytic disk with M; both estimates describe their precise domains without assuming convergence of the original infinite transseries.

6 Sharp late coefficients of a resonant Lambert inverse

The newest incoming Lambert manuscript leaves a precise question about the large-order coefficients of its inverse, under the heading "Sharp large-order constants of the inverse sectors" [8]. We prove its proposed even and odd constants, obtain a further relative correction, and identify the corresponding logarithm and dilogarithm terms inside the initial Borel disk. The proof also explains which changes to the analytic core or to the finite forward prefix can affect those terms. General algebras of factorially divergent series provide a broader setting for such questions [2]; the argument below supplies direct estimates for the specific coefficient extraction and its full sum over the number of factors. Lagrange inversion is used in its usual formal coefficient sense [1].

6.1 The source equation and its normalization

Set $A = 4\pi^2$ and consider the double Lambert sum

$$\mathcal{F}(t) = \sum_{a,b\ge 1} \frac{e^{-abt}}{ab} = \sum_{n\ge 1} \frac{d(n)}{n} e^{-nt}, \qquad t > 0.$$

Here $d(n)$ counts the positive divisors of n. We use the Stieltjes convention $\zeta(1+w) = w^{-1} + \sum_{j\ge 0}(-1)^j \gamma_j w^j/j!$. Its Mellin integrand is $\Gamma(w)\zeta(1+w)^2 t^{-w}$. The triple pole at zero gives the core

$$\frac{1}{2}(\log(1/t) + \gamma)^2 + C_*, \qquad C_* = \frac{\zeta(2)}{2} - \gamma^2 - 2\gamma_1.$$

The negative integer residues give the formal tail

$$R(q) = \sum_{n\ge 1} r_n q^n, \qquad r_1 = -\frac{1}{4}, \qquad r_{2m} = \frac{4\Gamma(2m)}{(2m)A^{2m}}\zeta(2m)^2, \qquad r_{2m+1} = 0 \quad (m \ge 1). \tag{6.1}$$

In particular $r_2 = 1/288$. The real completion of this expansion is treated in the predecessor [8]. Here we need only the explicit formal coefficients in (6.1).

For Y large, the exact-core variables of that manuscript are $X = \sqrt{2(Y - C_*)}$, $u = X^{-1}$, $q = e^{\gamma - X}$, and $t = qe^{-V}$. Their formal inverse equation is

$$V + \frac{u}{2}V^2 + uR(qe^{-V}) = 0, \qquad V(q,u) = \sum_{n\ge 1} C_n(u) q^n. \tag{6.2}$$

Throughout the coefficient analysis, q and u are independent; the physical specialization is a later operation. The formal implicit function theorem gives a unique solution. The predecessor proves that each C_n is a polynomial and obtains

$$C_n(u) = -\sum_{k=1}^{n} \frac{u^k}{k!}[q^n]R(q)^k \sum_{j=0}^{k-1} \binom{k-1}{j} n^{k-1-j}(k)_j \left(\frac{u}{2}\right)^j. \tag{6.3}$$

Its conjecture is that the dominant configurations contain one large even index and $k-1$ linear indices. The issue left open there is the uniform justification of summing that approximation over k.

Theorem 6.1 (Sharp Lambert inverse amplitudes)**.** *For every $U > 0$, uniformly for $|u| \le U$,*

$$C_n(u) = -\frac{2u\Gamma(n)}{nA^n}\left(e^{-Au/4} + (-1)^n e^{Au/4}\right) + O_U\left(\frac{\Gamma(n)}{n^2A^n}\right). \tag{6.4}$$

Consequently, for every fixed real $u > 0$,

$$C_{2m}(u) \sim -\frac{4u\Gamma(2m)}{(2m)A^{2m}}\cosh(Au/4), \tag{6.5}$$

$$C_{2m+1}(u) \sim \frac{4u\Gamma(2m+1)}{(2m+1)A^{2m+1}}\sinh(Au/4). \tag{6.6}$$

These are the two conjectured constants of the incoming manuscript.

The additive formulation (6.4) also covers complex values at which one of the displayed hyperbolic functions vanishes. It is stronger than a separate positive-real ratio asymptotic.

6.2 Analytic-core universality

Theorem 6.2 (Universality of the leading action amplitudes)**.** *Let $A > 0$, $r_1 = c$, and suppose*

$$r_n = \frac{\Gamma(n)}{nA^n}(\sigma_+ + (-1)^n\sigma_-) + O\left(\frac{\Gamma(n)}{n^2A^n}\right), \qquad n \ge 2. \tag{6.7}$$

Let $\Psi(v,u)$ be analytic near $v = 0$ and u in an open parameter domain, with $\Psi(0,u) = 0$ and $\partial_v\Psi(0,u) = 1$; assume a common neighborhood of $v = 0$ on each compact parameter set. The unique formal solution

$$\Psi(V,u) + uR(qe^{-V}) = 0, \qquad V(q,u) = \sum_{n\ge1} C_n(u)q^n,$$

satisfies, locally uniformly in u,

$$C_n(u) = -\frac{u\Gamma(n)}{nA^n}\left(\sigma_+e^{Acu} + (-1)^n\sigma_-e^{-Acu}\right) + O\left(\frac{\Gamma(n)}{n^2A^n}\right). \tag{6.8}$$

Thus nonlinear changes to the analytic core and finite changes to $r_2, r_3, \dots$ preserve the leading amplitudes. Changing $r_1 = c$ multiplies the two amplitudes by the corresponding exponential factors.

Lemma 6.3 (Factorial convolutions)**.** *For integers $N \ge j \ge 1$,*

$$\sum_{\substack{\ell_1+\cdots+\ell_j=N\\ \ell_i\ge1}} \prod_{i=1}^{j}(\ell_i-1)! \le 3^{j-1}(N-j)!. \tag{6.9}$$

Suppose $A|r_1| \le K$ and $A^m|r_m| \le K(m-2)!$ for $m \ge 2$. For $1 \le k < n$, with $N = n-k$,

$$A^n|[q^n]R(q)^k| \le (4K)^k\Gamma(N). \tag{6.10}$$

Moreover

$$[q^n]R(q)^k = kc^{k-1}r_{n-k+1} + E_{n,k}, \qquad A^n|E_{n,k}| \le (4K)^k\Gamma(N-1) \quad (N \ge 2). \tag{6.11}$$

Proof. The elementary inequality

$$\sum_{p=0}^{M} p!(M-p)! \le 3M!$$

holds because the two endpoint reciprocal binomial coefficients sum to 2, while each interior one is at most $1/M$; the cases $M = 0, 1$ are immediate. Induction gives (6.9).

In a composition $m_1 + \cdots + m_k = n$ put $\ell_i = m_i - 1$ and let j be the number of positive ℓ_i. The absolute sum is at most

$$K^k \sum_{j=1}^{\min(k,N)} \binom{k}{j} 3^{j-1}(N-j)!.$$

Bounding each factorial by $(N-1)!$ proves (6.10). The terms with $j = 1$ give exactly $kc^{k-1}r_{n-k+1}$; for the remaining $j \ge 2$ one may instead bound by $(N-2)!$, proving (6.11). □

Proof of Theorem 6.2. Write $G(v, u) = v/\Psi(v, u)$, analytically continued at $v = 0$, so that $G(0, u) = 1$. Lagrange inversion in an auxiliary multiplier gives

$$C_n(u) = -\sum_{k=1}^{n} \frac{u^k}{k!} n^{k-1} [q^n] R^k H_{n,k}(u), \tag{6.12}$$

where

$$\begin{aligned} H_{n,k}(u) &= (-n)^{1-k}\, \partial_v^{k-1}\{e^{-nv} G(v,u)^k\}\Big|_{v=0} \\ &= \sum_{j=0}^{k-1} \binom{k-1}{j} (-n)^{-j}\, \partial_v^j G(v,u)^k\Big|_{v=0}. \end{aligned} \tag{6.13}$$

Only $k \le n$ can contribute, so the auxiliary multiplier may be set to one formally. This also proves the coefficient formula without assuming convergence of R.

Fix a compact parameter set. Choose $\rho > 0$ and $M \ge 1$ such that $|G(v, u)| \le M$ on $|v| \le \rho$. Cauchy's estimate yields $|\partial_v^j G^k(0, u)| \le j! M^k \rho^{-j}$. Since $j! \le n^j$ for $j \le n$, there is a constant B with $|H_{n,k}| \le B^k$. The sharper inequality $j! \le j n^{j-1}$ for $j \ge 1$ gives

$$|H_{n,k} - 1| \le \frac{M^k (k-1) \rho^{-1} (1 + \rho^{-1})^{k-2}}{n} \le \frac{k B_1^k}{n}. \tag{6.14}$$

Here and below constants may depend on the compact parameter set.

Assumption (6.7) supplies the constant K of Lemma 6.3. After division by $\Gamma(n)/(nA^n)$, the absolute kth summand of (6.12), for $k < n$, is bounded by

$$\frac{D^k}{k!} \frac{n^k \Gamma(n-k)}{\Gamma(n)}. \tag{6.15}$$

For $k \le n/2$ the last ratio is at most 2^k, since it is the product of $n/(n-j)$ for $1 \le j \le k$. This produces the summable majorant $(2D)^k/k!$.

For $n/2 < k < n$, use $\Gamma(n-k) \le \Gamma(\lceil n/2 \rceil)$ and $n^k/k! \le (en/k)^k \le (2e)^k$. The entire contribution of this range is bounded by

$$n \max(1, 2eD)^n \frac{\Gamma(\lceil n/2 \rceil)}{\Gamma(n)}, \tag{6.16}$$

which is smaller than every inverse power of n. For $k = n$ one uses $[q^n]R^n = c^n$ directly; the normalized bound is a constant to the nth power divided by $\Gamma(n)$ and has the same property.

It remains to sharpen the range $k \le n/2$. By (6.11), the normalized total contribution of all $E_{n,k}$ is $O(1/n)$: indeed its kth bound differs from (6.15) by the extra factor $(n-k-1)^{-1} = O(1/n)$. For the remaining term put $m = n-k+1$. The gamma ratio needed in its evaluation is

$$B_{n,k} := \frac{n^k \Gamma(n-k+1)}{(n-k+1)\Gamma(n)} = \frac{n}{n-k+1} \prod_{j=1}^{k-1} \frac{n}{n-j}. \tag{6.17}$$

For $k \le n/2$ it is at most 2^k, and $|B_{n,k} - 1| \le Ck^2 2^k/n$. To see the latter, take the logarithm of the finite product, use $-\log(1-x) \le 2x$ for $0 \le x \le 1/2$, and then $e^y - 1 \le ye^y$.

The error in (6.7), with $m \ge n/2$, is also $O(1/n)$ after this normalization. Combining it with (6.14) and (6.17) shows that the kth term may be replaced, with a total summable error $O(1/n)$, by

$$-\frac{u(Acu)^{k-1}}{(k-1)!}(\sigma_+ + (-1)^{n-k+1}\sigma_-).$$

All error bounds have a polynomial in k times a constant to the kth power divided by $(k-1)!$. Summing the two exponential series proves (6.8), uniformly on the compact parameter set. □

Proof of Theorem 6.1. In (6.1), $\zeta(n)^2 = 1 + O(2^{-n})$ for $n \ge 2$. Thus (6.7) holds with $c = -1/4$ and $\sigma_+ = \sigma_- = 2$. Apply Theorem 6.2 to $\Psi(v,u) = v + uv^2/2$. The identities for cosh and sinh give (6.5)–(6.6). □

6.3 The first correction and the analytic data it detects

Theorem 6.4 (First relative correction)**.** *For the Lambert inverse (6.2), put*

$$\theta = Au/4, \qquad P(u) = \frac{1+u}{2}\theta^2 + \frac{uA^2}{288}, \qquad D_n(u) = \frac{nA^n C_n(u)}{4\Gamma(n)}. \tag{6.18}$$

Let $E_n(\theta) = \cosh\theta$ for even n and $E_n(\theta) = -\sinh\theta$ for odd n. Uniformly for $|u| \le U$,

$$D_n(u) = -uE_n(\theta) - \frac{u}{n}\left(P(u)E_n(\theta) + (u+2)\theta E_n'(\theta)\right) + O_U(n^{-2}). \tag{6.19}$$

The correction detects the quadratic core and $r_2 = 1/288$, although neither affects the leading amplitudes.

Proof. We retain one more order in the preceding proof. Uniformly for $n \ge 8$ and $k \le n/2$, with $\ell = k-1$, expansion of the logarithm of (6.17) gives

$$B_{n,k} = 1 + \frac{\ell(\ell+3)}{2n} + O\left(\frac{k^4 B_2^k}{n^2}\right). \tag{6.20}$$

For example, the sum of the first-order logarithmic terms is $\ell(\ell+3)/(2n)$; the sum of their squared arguments is $O(k^3/n^2)$, and the exponential remainder is bounded by $O(k^4 B_2^k/n^2)$.

For the quadratic core, (6.13) similarly gives

$$H_{n,k} = 1 + \frac{uk(k-1)}{2n} + O_U\left(\frac{k^4 B_3^k}{n^2}\right). \tag{6.21}$$

This is a uniform estimate, not just a fixed-k expansion. In the Cauchy argument, the terms $j \ge 2$ are bounded using $j! \le j(j-1)n^{j-2}$; their binomial sum is bounded by a polynomial in k times a constant to the kth power, divided by n^2. For a general core $\Psi(v,u) = v + b(u)v^2 + O(v^3)$, the first correction in (6.21) is $b(u)k(k-1)/n$.

The one-large-index configurations therefore contribute the operator

$$\frac{1}{2}\Big((1+u)(\theta\partial_\theta)^2+(3+u)\theta\partial_\theta\Big)$$

applied to the parity exponential sum. Since $E_n''=E_n$, its value is

$$\frac{1+u}{2}\theta^2E_n+(u+2)\theta E_n'. \tag{6.22}$$

There is one further contribution of normalized order $1/n$: exactly two indices are at least two, one equals 2, and the other equals $n-k$. For $k\le n/2$ and sufficiently large n, its coefficient is exactly

$$k(k-1)c^{k-2}r_2r_{n-k}.$$

Using its leading gamma ratio and summing over k gives $-A^2r_2u^2E_n/n$. Together with (6.22), this is precisely the correction in (6.19).

For completeness, the configurations omitted in this refinement have a uniform $O(n^{-2})$ bound after normalization. Those with $j\ge 3$ indices at least two satisfy the strengthened bound $(4K)^k\Gamma(n-k-2)$ in Lemma 6.3. With $k\le n/2$ the two additional inverse factors are $O(n^{-2})$. For $j=2$, remove the two endpoint configurations just used. If $N=n-k$, both remaining increments are at least two, and

$$\sum_{\ell=2}^{N-2}(\ell-1)!(N-\ell-1)!\le 4(N-3)!. \tag{6.23}$$

Indeed, put $M=N-2$, divide by $M!$, and use $\sum_{i=1}^{M-1}\binom{M}{i}^{-1}\le 4/M$: the two endpoints contribute $2/M$, while the remaining binomial coefficients are at least $M(M-1)/2$. Again two inverse powers of n result. The gamma and derivative remainders in the retained configurations have the same bound after summing their factorial majorants. The range $k>n/2$ is smaller than every inverse power by (6.16). This proves the stated uniform remainder. □

6.4 Exact logarithm–dilogarithm decomposition in the Borel disk

The coefficient asymptotics have a direct polylogarithmic interpretation. It only requires summing absolutely convergent series in the initial disk and consequently does not require a continuation theorem for the inverse Borel transform.

Corollary 6.5 (The two nearest Borel singularities)**.** *For*

$$\widehat{V}_u(\xi)=\sum_{n\ge1}\frac{C_n(u)}{\Gamma(n)}\xi^{n-1}$$

there is, on $|\xi|<A$, the exact decomposition

$$\widehat{V}_u(\xi)=\frac{2u}{\xi}\Big(e^{-\theta}\log(1-\xi/A)+e^{\theta}\log(1+\xi/A)\Big)+J_u(\xi), \tag{6.24}$$

where J_u is holomorphic in the disk and continuous on its closure, locally uniformly in u. The factor $1/\xi$ has a removable singularity at zero in the displayed combination. For every complex $u\neq0$, both endpoints $+A$ and $-A$ are logarithmically singular, with radial logarithmic coefficients respectively

$$\frac{2u}{A}e^{-\theta},\qquad -\frac{2u}{A}e^{\theta}. \tag{6.25}$$

The corresponding radially regularized limits exist. The Borel radius is exactly A, and the ordinary q-series has radius zero. At $u=0$ the formal solution is identically zero.

Proof. Divide (6.4) by $\Gamma(n)$ and sum using $\sum_{n\geq 1} z^n/n = -\log(1-z)$. The error series is absolutely and uniformly convergent on $|\xi| \leq A$, because its coefficients are bounded by a constant times A^{-n}/n^2. This proves (6.24). The coefficients in (6.25) never vanish for $u \neq 0$, proving both endpoint singularities and the exact Borel radius. At least one of $\cosh\theta$ and $\sinh\theta$ is nonzero, so (6.4) on that parity subsequence also proves that the ordinary radius is zero. For $u = 0$, formal uniqueness gives $V = 0$. □

Corollary 6.6 (An explicit dilogarithmic correction)**.** *With P from (6.18), set*

$$Q_+(u) = P(u) - (u+2)\theta, \qquad Q_-(u) = P(u) + (u+2)\theta. \tag{6.26}$$

The following identity holds throughout $|\xi| < A$:

$$\begin{aligned}\widehat{V}_u(\xi) = \frac{2u}{\xi}\left(e^{-\theta}\log(1-\xi/A) + e^{\theta}\log(1+\xi/A)\right)\\ - \frac{2u}{\xi}\left(e^{-\theta}Q_+(u)\,\mathrm{Li}_2(\xi/A) + e^{\theta}Q_-(u)\,\mathrm{Li}_2(-\xi/A)\right) + K_u(\xi).\end{aligned} \tag{6.27}$$

Both K_u and $\partial_\xi K_u$ extend continuously to the closed disk, locally uniformly in u.

Proof. Theorem 6.4 is equivalent to

$$\begin{aligned}\frac{C_n(u)}{\Gamma(n)} = &-\frac{2u}{nA^n}(e^{-\theta} + (-1)^n e^{\theta})\\ &-\frac{2u}{n^2A^n}(e^{-\theta}Q_+(u) + (-1)^n e^{\theta}Q_-(u)) + O_U(A^{-n}n^{-3}).\end{aligned}$$

Sum this identity with $\mathrm{Li}_2(z) = \sum_{n\geq 1} z^n/n^2$. The remainder and its once-differentiated series converge absolutely and uniformly on $|\xi| \leq A$, proving the assertion. □

These are statements from inside the initial Borel disk. They neither assert analytic continuation through that circle nor identify all subsequent singularities. They do distinguish the two leading endpoint amplitudes, improve the predecessor's radius-only result to every nonzero complex u, and express the first correction by explicit classical polylogarithms.

7 Exact ramification of the inverse at the polylogarithm endpoint

The endpoint of a forward function need not be a singular point of its inverse. For the polylogarithm this distinction produces an infinite discrete family of removable endpoints. We give an exact classification, including the smallest branched cover that resolves every noninteger rational order. The convergent nonintegral charts underlying the result already occur in Proposition `prop:noninteger` of the repository report [9]. Our contribution here is the minimal ramification degree, the resulting if-and-only-if removability criterion, and explicit Taylor data at the exceptional orders. The general use of Puiseux uniformization is classical.

Fix a real number $s > 1$, put $\rho_s = \zeta(s)$, and let G_s denote the inverse germ of Li_s continued from the interval $0 < w < \rho_s$. Since Li_s increases strictly from 0 to ρ_s on $0 < z < 1$, this specifies the branch without a complex sheet convention. Near its endpoint write

$$y = \rho_s - w, \qquad t = -\log G_s(\rho_s - y), \qquad \Delta_s(t) = \rho_s - \mathrm{Li}_s(e^{-t}).$$

Initially $y, t > 0$. All subsequent identities refer to the analytic continuations of this local branch. In particular, a continuation of G_s through ρ_s need not remain an inverse of the principal polylogarithm on both sides of its chosen cut.

We say that the endpoint has finite ramification degree d if d is the smallest positive integer for which $G_s(\rho_s - \eta^d)$ extends holomorphically to $\eta = 0$. The substitution is made on the branch agreeing with positive η. Degree one means that the endpoint is removable for the inverse. When no such integer exists, we say that the endpoint has infinite ramification. Equivalently, the bounded local inverse is not resolved by any finite branched cover; thus it cannot have finitely many local monodromy branches around the endpoint.

For nonintegral s, the classical convergent Jonquière expansion [4] gives

$$\Delta_s(t) = c_s t^\alpha + A_s(t), \qquad \alpha = s - 1, \quad c_s = -\Gamma(-\alpha), \tag{7.1}$$

where

$$A_s(t) = \sum_{k\geq 1} a_k(s) t^k, \qquad a_k(s) = \frac{(-1)^{k+1}}{k!}\zeta(s-k), \qquad |t| < 2\pi. \tag{7.2}$$

The power in (7.1) is taken on the selected logarithmic cover. Notice that $c_s > 0$ for $1 < s < 2$, whereas $a_1(s) = \zeta(s-1) > 0$ for $s > 2$.

Theorem 7.1 (Minimal endpoint ramification)**.** *Let $s > 1$ be real.*

1. *If s is nonintegral and $s - 1 = p/q$ in lowest positive terms, then the endpoint ramification degree of G_s is*
$$d_s = \min\{p, q\}.$$
2. *If $s - 1$ is irrational, the endpoint has infinite ramification.*
3. *If s is an integer with $s \geq 2$, the endpoint has infinite ramification.*

Consequently, G_s extends holomorphically through $w = \zeta(s)$ if and only if

$$s = 1 + \frac{1}{q}, \qquad q = 2, 3, 4, \ldots. \tag{7.3}$$

Proof. The local changes $z = e^{-t}$ and $t = -\log z$ are biholomorphic near $t = 0$ and $z = 1$. Thus G_s and $t(y)$ have exactly the same ramification degree.

First suppose $\alpha = p/q$ is rational and nonintegral. Put $t = v^q$. If $p < q$, equation (7.1) becomes

$$y = v^p B(v), \qquad B(v) = c_s + \sum_{k\geq 1} a_k(s) v^{qk-p}, \qquad B(0) = c_s \neq 0.$$

Choose the analytic pth root of B which is positive at the origin. The map $v \mapsto \eta = vB(v)^{1/p}$ has nonzero derivative there. Its holomorphic inverse $v = V(\eta)$ gives

$$y = \eta^p, \qquad t(y) = V(\eta)^q, \qquad G_s(\rho_s - \eta^p) = e^{-V(\eta)^q}.$$

This proves that degree p suffices. Conversely,

$$t(y) = c_s^{-q/p} y^{q/p}(1 + o(1)). \tag{7.4}$$

If $t(\eta^d)$ were holomorphic at zero, its order of vanishing would be dq/p, an integer. Coprimality forces $p \mid d$. Thus p is minimal.

If $p > q$, write instead

$$y = v^q B(v), \qquad B(v) = a_1(s) + c_s v^{p-q} + \sum_{k\geq 2} a_k(s) v^{q(k-1)}, \qquad B(0) > 0.$$

The map $\eta = vB(v)^{1/q}$ is again a local analytic coordinate, so that $y = \eta^q$ and $t = V(\eta)^q$. Hence degree q suffices. To prove minimality, let H_s be the analytic inverse of A_s at zero and put $a = a_1(s)$. Substitution into (7.1), followed by one Taylor expansion of A_s, gives

$$t(y) - H_s(y) = -c_s a^{-\alpha-1} y^\alpha (1 + o(1)), \qquad y \downarrow 0. \tag{7.5}$$

Indeed, $t \sim y/a$, the difference is $O(y^\alpha)$, and the linear term of $A_s(t) - A_s(H_s(y))$ is $a(t - H_s(y))(1 + o(1))$. Since $c_s \neq 0$, a holomorphic $t(\eta^d) - H_s(\eta^d)$ would have vanishing order $d\alpha = dp/q$. This forces $q \mid d$ and establishes minimality.

For irrational $0 < \alpha < 1$, the same leading law $t(y) = c_s^{-1/\alpha} y^{1/\alpha}(1 + o(1))$ rules out every finite degree d, because d/α is not an integer. For irrational $\alpha > 1$, equation (7.5) rules out every finite degree because $d\alpha$ is not an integer.

It remains to check integer orders. For $s = m \geq 3$, the classical resonant expansion has the form

$$\Delta_m(t) = A_m(t) + b_m t^{m-1} \log t, \qquad A_m'(0) = \zeta(m-1) > 0, \qquad b_m = \frac{(-1)^{m-1}}{(m-1)!} \neq 0, \tag{7.6}$$

with A_m analytic. If $H_m = A_m^{-1}$ and $a = \zeta(m-1)$, the preceding first-correction argument now yields

$$t(y) - H_m(y) = -b_m a^{-m} y^{m-1} \log y \, (1 + o(1)). \tag{7.7}$$

After $y = \eta^d$, its nonzero leading term is a constant times $\eta^{d(m-1)} \log \eta$. No holomorphic function has this leading behavior: its ratio to its first nonzero integral power is a finite nonzero constant. Therefore no finite degree suffices.

At $m = 2$,

$$\Delta_2(t) = t \log(1/t) + t + O(t^2), \qquad t(y) \sim \frac{y}{\log(1/y)}.$$

If $t(\eta^d)$ were holomorphic and nonzero, its leading monomial would be $C\eta^j$ for some integer $j \geq 1$ and $C \neq 0$. The displayed asymptotic would force $j = d$, but then the leading coefficient would be zero, a contradiction. This treats all cases. Since a nonintegral rational $\alpha > 1$ has $q \geq 2$, while $0 < \alpha < 1$ gives degree one precisely when $p = 1$, the final criterion follows. □

Remark 7.2 (Scope of the monodromy statement). For rational orders the proof explicitly constructs a holomorphic function on a finite cover, and its minimal degree is the exact number of local branches. In the other cases it proves an obstruction to every finite cover. If a bounded inverse had finite local monodromy, passing to its finite covering would make it single-valued and its bounded singularity removable, contradicting that obstruction. This is a local assertion. It neither identifies every singularity of the inverse continued globally nor determines the Taylor radius at the original point $w = 0$.

7.1 Taylor charts at the removable endpoints

At the exceptional order $s_q = 1 + 1/q$ define

$$\rho_q = \zeta(s_q), \qquad c_q = -\Gamma(-1/q) = q\Gamma(1 - 1/q) > 0,$$

and the convergent analytic germ

$$Y_q(v) = c_q v + \sum_{k \geq 1} \frac{(-1)^{k+1} \zeta(1 + 1/q - k)}{k!} \, v^{qk}. \tag{7.8}$$

Let $V_q = Y_q^{-1}$ be its analytic local inverse. Then the exact endpoint inverse is

$$G_{s_q}(\rho_q - y) = \exp(-V_q(y)^q). \tag{7.9}$$

The series defining Y_q converges whenever $|v|^q < 2\pi$; the inverse has a positive local radius by $Y_q'(0) = c_q$. No assertion that these two radii agree is needed.

Proposition 7.3 (Finite coefficient formula and arithmetic support)**.** *Put* $a_k = (-1)^{k+1}\zeta(1 + 1/q - k)/k!$ *and write*

$$G_{s_q}(\rho_q - y) = 1 + \sum_{n\geq 1} g_{q,n} y^n.$$

For every $n \geq 1$*,*

$$g_{q,n} = -\frac{q}{n}[v^{n-q}]e^{-v^q}\left(c_q + \sum_{k\geq 1} a_k v^{qk-1}\right)^{-n}, \tag{7.10}$$

where a negative coefficient index means zero. Every extraction is finite. The support satisfies

$$\{n \geq 1 : g_{q,n} \neq 0\} \subseteq q + \langle q-1, q\rangle = \{q + j(q-1) + kq : j, k \geq 0\}. \tag{7.11}$$

In particular,

$$g_{q,q} = -c_q^{-q}, \qquad g_{q,2q-1} = \frac{q\zeta(1/q)}{c_q^{2q}} \neq 0, \tag{7.12}$$

and $g_{q,n} = 0$ *for* $q < n < 2q - 1$*.*

Proof. Lagrange–Bürmann applied to $e^{-V_q(y)^q}$ gives

$$[y^n]e^{-V_q(y)^q} = \frac{1}{n}[v^{n-1}](-qv^{q-1}e^{-v^q})\left(\frac{v}{Y_q(v)}\right)^n,$$

which is (7.10). In its last product, the available positive exponents are q and $qk - 1 = (q-1) + (k-1)q$. Their sums give (7.11). Extracting the constant term and then the term of degree $q - 1$ proves (7.12). Finally $\zeta(1/q) < 0$ because the alternating Dirichlet eta series is positive and $1 - 2^{1-1/q} < 0$. □

For $q = 2$, set $a = \zeta(1/2)$. Since $c_2 = 2\sqrt{\pi}$, the first Taylor terms are

$$G_{3/2}(\zeta(3/2) - y) = 1 - \frac{y^2}{4\pi} + \frac{a\,y^3}{8\pi^2} + \left(\frac{1}{32\pi^2} - \frac{5a^2}{64\pi^3}\right) y^4 + O(y^5). \tag{7.13}$$

Thus a square-root singularity of the forward function produces a holomorphic inverse with a quadratic critical point.

For $q = 3$, put $c = 3\Gamma(2/3)$ and $a = \zeta(1/3)$. Then

$$G_{4/3}(\zeta(4/3) - y) = 1 - \frac{y^3}{c^3} + \frac{3a\,y^5}{c^6} + \frac{y^6}{2c^6} - \frac{12a^2y^7}{c^9} + O(y^8). \tag{7.14}$$

The missing fourth-degree term is a structural gap, explained by (7.11). For general q, derivatives of orders $1, \ldots, q-1$ vanish at the endpoint and

$$G_{s_q}^{(q)}(\rho_q) = (-1)^{q+1}\frac{q!}{c_q^q} \neq 0.$$

7.2 A correction to a proposed endpoint-circle extension

The repository's integer-order cut-modulus theorem [9] states that $|\operatorname{Li}_s(e^u + i0)| > \zeta(s)$ for integer $s \geq 2$ and $u > 0$. Its further-research question 9 asks for an all-real-order endpoint-circle theorem and points to the integer cut-modulus argument. The corresponding all-real extension of this cut-modulus ingredient fails below $s = 3/2$. This does not contradict the proved integer theorem or by itself settle the full endpoint-circle question.

Proposition 7.4 (Sharp local threshold for the cut-modulus inequality)**.** *For fixed $1 < s < 3/2$, one has*

$$|\operatorname{Li}_s(e^u + i0)| < \zeta(s)$$

for all sufficiently small $u > 0$. For every fixed $s \geq 3/2$, the opposite strict inequality holds for all sufficiently small $u > 0$. This proposition concerns a neighborhood of the endpoint, rather than the entire cut.

Proof. For $1 < s < 2$, put $\alpha = s - 1 \in (0, 1)$ and $\rho = \zeta(s)$. On the upper bank $t = -u - i0$, so Jonquière's formula gives

$$\operatorname{Li}_s(e^u + i0) = \rho + \Gamma(-\alpha)e^{-i\pi\alpha}u^\alpha + \zeta(\alpha)u + O(u^2).$$

If $\alpha \neq 1/2$, then

$$|\operatorname{Li}_s(e^u + i0)|^2 - \rho^2 = 2\rho\Gamma(-\alpha)\cos(\pi\alpha)u^\alpha + o(u^\alpha).$$

Since $\Gamma(-\alpha) < 0$, its sign is negative for $0 < \alpha < 1/2$ and positive for $1/2 < \alpha < 1$. At $\alpha = 1/2$, the leading fractional term is purely imaginary, and the next calculation gives

$$|\operatorname{Li}_{3/2}(e^u + i0)|^2 - \zeta(3/2)^2 = (4\pi + 2\zeta(3/2)\zeta(1/2))u + O(u^2). \tag{7.15}$$

The coefficient is strictly positive. For an elementary certificate, $\zeta(3/2) < 3$ follows by integral comparison, and $-2 < \zeta(1/2) < 0$ follows from the fifth alternating partial sum

$$0 < \eta(1/2) < 1 - 2^{-1/2} + 3^{-1/2} - \tfrac{1}{2} + 5^{-1/2} < 2(\sqrt{2} - 1).$$

The last strict inequality may be checked by elementary squaring; for example the rational bounds $\sqrt{2} > 1414/1000$, $1/\sqrt{2} > 707/1000$, $1/\sqrt{3} < 578/1000$, and $1/\sqrt{5} < 448/1000$ already prove it. Consequently the coefficient in (7.15) exceeds $4\pi - 12 > 0$. For $s > 2$, the boundary increment is $\zeta(s-1)u + o(u)$ with positive real coefficient, and for $s = 2$ it is $u(1 - \log u) + i\pi u + O(u^2)$. These establish the remaining local cases. □

For example, at $s = 5/4$ and $u = 10^{-6}$, a 60-digit evaluation gives $|\operatorname{Li}_{5/4}(e^u + i0)| - \zeta(5/4) \approx -0.108266505478287898$. The sign is proved by the preceding asymptotic argument; the decimal is only a diagnostic. The modulus gap concerns the boundary of the principal-sheet image. A regular point on an image cut can be crossed by analytic continuation of the inverse. Therefore the counterexample alone does not locate the true nearest singularity of the inverse germ at $w = 0$.

8 Verification, interpretation, and integration

The proofs establish the infinite statements. The computations detect normalization errors, compare independent finite constructions, and illustrate the size of the proved remainders. Machine-readable outputs are retained beside the programs that generated them.

8.1 Exact finite checks

The angular program verifies the first four universal degree polynomials by symbolic coefficient extraction and supplies the rational certificate for the derivative constant in Theorem 5.1. The endpoint program compares fixed-point formal reversion with the independent Lagrange formula (7.10); fifteen coefficients for $q = 2, 3$ agree exactly. Its zero coefficients also agree with the support restriction.

The Lambert program evaluates every factor count $1 \le k \le n$ in (6.3), including all convolution powers through $n = 201$. The powers use the exact forward coefficient formula at high precision, without the large-order approximation. The cases include two positive real parameters, a generic complex parameter, and $u = i/(2\pi)$, where $\cosh(Au/4) = 0$. That last case tests why the complex-parameter result must be additive rather than a ratio to a possibly vanishing leading term.

8.2 Angular numerical diagnostics

For positive integral outer order, an independent evaluation is

$$F_{a,b}(z) = \frac{1}{\Gamma(a)} \int_0^\infty t^{a-1} \operatorname{Li}_0(ze^{-t}) \operatorname{Li}_b(ze^{-t})\, dt. \tag{8.1}$$

It follows by coefficientwise integration in the disk and continuation. The program solves the angular equation using this integral and compares the result with coefficients formed from (4.1). High derivatives at the boundary use analytic lowering, including (4.4), rather than a differentiated Fourier sum outside its convergence range. The integral uses a precision-dependent finite upper endpoint. Its exponentially small omitted tail is far below the displayed digits in these diagnostics; this implementation is not an interval integrator.

At $b = 1$, $\rho = 1$, the displacements include

$$\begin{aligned}
\delta_{6,1}(1) &= 0.06441843932204966951480818053626053968\ldots,\\
\delta_{8,1}(1) &= 0.02892818590626336166005258862627886175\ldots,\\
\delta_{10,1}(1) &= 0.01293779047910625463765885200102238078\ldots.
\end{aligned}$$

The bound is designed to work simultaneously for all $b > 0$ and all radii, and is conservative near the lowest guaranteed outer order. Table 1 displays the numerical error and the analytic bound separately.

Table 1: Degree reconstruction at $b = 1$, $\rho = 1$. Errors are 85-digit numerical diagnostics; bounds evaluate (3.5).

a	K	Numerical error	Proved upper bound
6	4	1.15×10^{-5}	2.14×10^{-1}
6	8	3.06×10^{-8}	7.05×10^{-2}
6	16	9.24×10^{-15}	1.32×10^{-2}
8	4	4.42×10^{-7}	7.42×10^{-4}
8	8	8.08×10^{-12}	9.55×10^{-6}
8	16	4.19×10^{-19}	2.72×10^{-9}
10	4	9.79×10^{-9}	9.49×10^{-6}
10	8	2.23×10^{-14}	4.77×10^{-9}
10	16	1.64×10^{-26}	2.06×10^{-15}

For the moving construction with $M = \lfloor (3/2)^{a/2} \rfloor$ and $K = a$, the comparison is shown in Table 2. The finite expression at $a = 28$ has an observed error below $4 \cdot 10^{-61}$, while the theorem gives the more conservative analytic bound in the last column.

Table 2: Finite frequency-and-degree approximations at $b = 1$, $\rho = 1$. Bounds evaluate (5.3).

a	M	K	Numerical error	Proved upper bound
20	57	20	6.05×10^{-30}	1.06×10^{-16}
24	129	24	4.10×10^{-44}	2.01×10^{-28}
28	291	28	3.88×10^{-61}	6.30×10^{-43}

8.3 Lambert and endpoint diagnostics

Table 3 uses $D_n = nA^nC_n/(4\Gamma(n))$ and Theorem 6.4. Including the first correction decreases the observed error by a further factor on the order of n. The full complex-parameter record is retained in the JSON output.

Table 3: Lambert coefficient diagnostics at $u = 0.1$. The errors compare the full finite formula with the leading term, and with the leading term plus the first correction.

n	Leading-term error	Error with first correction
40	1.176×10^{-2}	1.657×10^{-3}
41	1.228×10^{-2}	1.525×10^{-3}
80	5.430×10^{-3}	3.801×10^{-4}
81	5.804×10^{-3}	3.588×10^{-4}
200	2.078×10^{-3}	5.799×10^{-5}
201	2.250×10^{-3}	5.556×10^{-5}

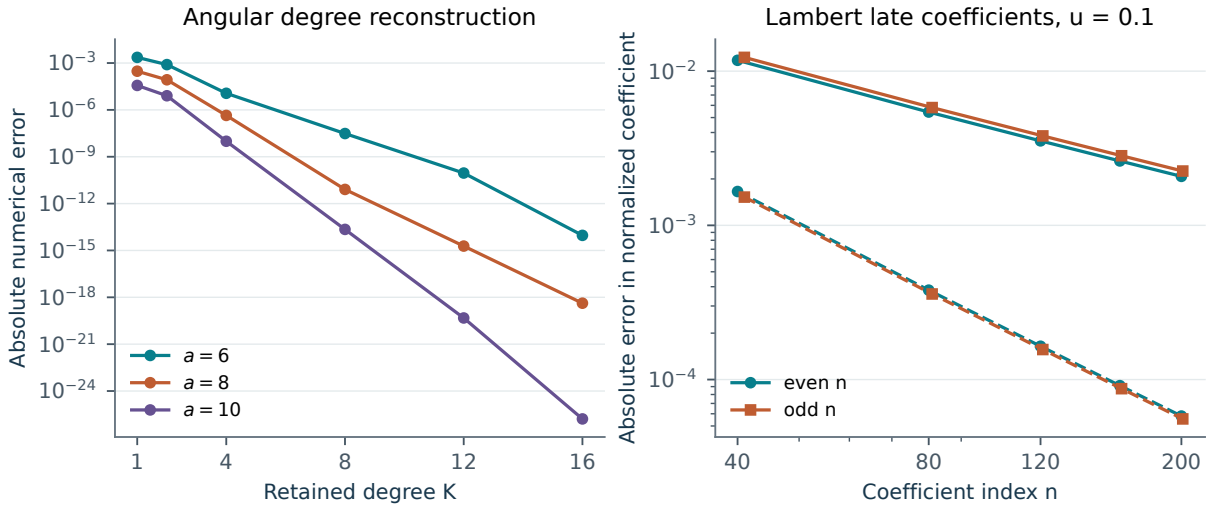


Figure 1: Numerical consequences of the proofs. Left: angular degree reconstruction at $b = 1$, $\rho = 1$. Right: Lambert errors at $u = 0.1$, with even and odd indices separated. Solid curves use the leading term; dashed curves include the first correction. Points are diagnostics, not interval certificates.

For the removable endpoint expansions at $y = 10^{-3}$, direct polylogarithm root calculations give polynomial errors approximately $1.7191 \cdot 10^{-19}$ for (7.13) and $1.0461 \cdot 10^{-29}$ for (7.14). Four cut-bank tests compare direct evaluation with an independently formed Jonquière expansion. The local threshold is proved using exact sign estimates; the decimals only check normalization and branch conventions.

8.4 Recommended changes to the existing material

No incorrect theorem was found in the particular angular finite-cutoff, angular divergence, integer cut-modulus, or Lambert coefficient statements used here. The justified integration changes are:

1. Mark the angular summation question as answered for $a \geq 6$, retaining the nonlinear-degree and radial-degree data. Preserve the source's divergence warning.
2. Add the finite approximation as progress on moving truncation, while keeping sharp optimality open.
3. Replace the two named Lambert conjectures with Theorem 6.1; cross-reference the first correction and Borel decomposition. Extend the exact-radius assertion in this model to every complex $u \neq 0$.
4. Add the minimal direct-inverse ramification classification as a refinement of the existing noninteger charts.
5. Record the failure of the all-real cut-modulus extension for $1 < s < 3/2$ and the sharp local threshold. Retain the integer-order theorem; leave both the global noninteger cut-modulus range and the broader endpoint-circle question open.

These are changes of mathematical status and scope. Correct historical reports need not be rewritten, and an open question should not be described as an erroneous theorem.

9 Further research questions

9.1 Angular summation and finite approximation

1. Find the least uniform outer-order threshold. Determine the least a_* for convergence of the scalar coupling Taylor series at $\tau = 1$, uniformly in $b > 0$ and $0 < \rho \leq 1$. The explicit $a \geq 6$ uses a simple disk and rational constants. A sharper norm may still miss the true obstruction: a complex coupling singularity satisfying $v = \tau f(v)$ and $1 = \tau f'(v)$, or a collision with the polylogarithm cut.

2. Prove a sharp optimal truncation law. Theorem 5.2 attains every $\lambda < \log(3/2)$ in its Gaussian exponent for one double cutoff. Determine the sharp error for the original single exponential-base cutoff and compare the two schemes. A lower bound must account for the prime-frequency divergence mechanism.

3. Compare natural summation prescriptions. Compare analytic continuation in the scalar coupling, radial Abel degree summation, and limits of exact finite-Fourier roots. Agreement must be proved in a common domain. A summation method depending only on coefficients already grouped by exponential base is a separate problem because the present construction retains additional multiset data.

4. Reduce the fixed period algebra. Use the repository's cyclotomic and depth-reduction identities to reduce the generating set in (4.5). This asks for proved reductions, not numerical independence. Compact generator lists could simplify many consecutive coupling coefficients.

5. Extend to other colors and depths. The local construction needs a simple dominant zero and a controlled complex tail. Identifying it with the intended global geometric zero requires

a separate global theorem. Keep those tasks distinct when studying other root-of-unity colors or higher-depth harmonic sums.

9.2 Factorial reversion and Borel geometry

6. Determine the full inverse Borel continuation. Does the inverse Borel germ continue along every path avoiding an explicit additive action set? The log–dilog decomposition only describes the initial disk and its boundary regularity. A continuation theorem should separate inherited arithmetic singularities from those created by nonlinear convolution.

7. Derive all relative late terms. The leading amplitudes detect r_1; the next term also detects r_2 and the quadratic core. Prove exactly which finite forward and core data determine each coefficient of n^{-j} and give a recursive formula. Borinsky's framework [2] is a natural starting point, with the two-action parity kept explicit. Successive terms should lead to higher polylogarithms in a refined initial-disk decomposition.

8. Resolve physical moving-parameter limits. The physical inverse has $u = 1/X$ and $q = e^{\gamma - X}$. A useful optimal-truncation analysis must choose n jointly with X. Derive the transition between the action amplitudes as $u \to 0$ and quantify the smallest remainder on that physical curve.

9. Study cancellation of a parity amplitude. At imaginary couplings an even or odd leading amplitude can vanish. Classify the first nonzero late term on each such locus. The first correction is a starting point; the cancellation of one parity amplitude does not remove either individual logarithmic action.

9.3 Global inverse geometry

10. Determine the global cut-modulus range. The threshold $s = 3/2$ is exact locally. For which noninteger real $s \geq 3/2$ does the strict inequality persist for every $u > 0$? Control the entire cut and distinguish the chosen principal-sheet image boundary from intrinsic inverse singularities.

11. Locate the nearest singularities at exceptional orders. At $s = 1 + 1/q$, $\zeta(s)$ is regular for the continued inverse, with a critical point of order q. What controls the Taylor radius of the original germ at $w = 0$? Candidates include critical values on other sheets and finite asymptotic values. The local removability theorem does not choose among them.

12. Compare direct and feedback ramification. Seek a local classification for $\Psi(v) = \mathrm{Li}_s(\Phi(v))$ with prescribed vanishing orders of Ψ and $\Phi - 1$. Quadratic feedback can change the direct degree $\min(p, q)$ and create folds. Resonant logarithms must be treated separately from rational powers; deleting them can produce a false finite-cover claim.

13. Formalize the reusable finite algebra. The finite factorial convolution inequalities, coefficient extraction, parity formulas, and rational norm certificates are suitable first formalization targets. Full analytic formalization also needs the polylogarithm continuation and kernel machinery, and the relevant complex-analytic theorems. This package makes no claim that those prerequisites or the present proofs have already been formalized.

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